

# Exercise Sheet 9

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## Diffusion and Epidemics

5942UE Network Science

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# 9.A Diffusion as a Network Process

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# Exercise 9.A.1: Simple Diffusion Trace

Directed graph:                   ,                   ,                   ,                   .

Node    is active at           . At each step, an active node activates all out-neighbours.

1. Trace the active set at           .
2. Which node is never reached? Why?
3. If you add the edge           , does anything change for this seed?

# Exercise 9.A.2: Structural Bottlenecks

Two 8-node graphs with seed :

- **Star:** connected to every other node.
  - **Line:**
1. How many steps until all nodes are active in each?
  2. Where is the structural bottleneck?
  3. A single edge fails. Which topology is more affected?

# Exercise 9.A.3: Centrality and Diffusion Speed

Load the karate club graph and BFS-diffuse from the highest-degree node.

1. How many steps until all 34 nodes are active?
2. Repeat from the lowest-degree node. How does the curve change?
3. What does this say about hubs?

## Exercise 9.A.4: Paths and Reachability

Company-wide email cascade from CEO ( ) must reach 500 employees. The network has a core-periphery structure: 50 managers in a dense core, each linked to        peripheral employees.

1. Does a message from        to a peripheral employee require passing through the core?
2. If 5 of 50 managers are absent, what fraction of periphery is potentially unreachable?
3. How could the company become more resilient?

# 9.B Epidemic Models and Simple Contagion

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## Exercise 9.B.1: SIR Trace

SIR rules (synchronous):            if any infected neighbour, with probability    .            at the next step (single-step infectious period).

Path graph            with    infected at            ,            .

1. Trace each node's state at            .
2. When does    become infected?
3. With            , what is            ?

# Exercise 9.B.2: Structural Effect of Topology

SIR with , one-step recovery, seed = node 1.

- **Line L:**
  - **Star S:** node 1 is the hub of
1. Maximum simultaneously infected in L?
  2. Maximum simultaneously infected in S?
  3. Which produces a sharper epidemic peak? Implications?

# Exercise 9.B.3: SIR Simulation in NetworkX

Implement synchronous SIR on the karate club graph and analyse it.

1. At what step does the epidemic peak?
2. What fraction ends in  $I$  (ever infected)?
3. Vary  $\beta$  from 0.1 to 0.8. Where does the epidemic reliably infect  $I$ ?

## Exercise 9.B.4: Simple vs. Complex Contagion

For each scenario, decide whether it is **simple** (one contact sufficient) or **complex** (multiple confirmations needed):

(a) A cold virus. (b) Adopting a new software tool only after three colleagues recommend it. (c) Sharing a news article posted by one friend. (d) Joining a protest after pressure from multiple community members. (e) Updating a belief after seeing a claim from several independent sources.

For (b) and (d), how must network structure differ from simple contagion?

# Wrap-Up

- diffusion is bounded by reachable paths; redundant paths don't help unless primary paths fail
- bottlenecks and hubs control epidemic dynamics
- the same produces dramatically different curves on different topologies
- complex contagion reverses the role of bridges and clusters compared to simple contagion